

CMPUT 328 — Assignment 1

Logistic Regression and Fully Connected Neural Networks

Total weight	9% of the course grade — Part 1 (MNIST) 4% + Part 2 (CIFAR-10) 5%
Deadline	21 September 2026, 11:59 PM (MST)
Submission	1. Completed Colab notebook with all required code, outputs, and plots visible. 2. A 1–2 page PDF report. Include your name and student ID on both files. If AI assistance is used, document it in the report.

Goal

Build and train classification models on MNIST and CIFAR-10, and analyze their training dynamics, evaluation results, and errors. You will work with logistic regression and fully connected neural networks, and investigate the effects of regularization, optimizers, training data, and model changes.

Deliverables

- Completed Colab notebook with code, outputs, plots, and short comments.
- A 1–2 page PDF report summarizing your findings.
- Include your name and student ID on both files.
- If you use Gemini or another AI assistant, document your usage in the report.

Part 1 (4%) — Logistic Regression on MNIST

Implement and train Logistic Regression on MNIST.

1. Baseline Logistic Regression

Train a baseline Logistic Regression model using the SGD optimizer and Cross-Entropy loss.

2. Regularization

Investigate the effect of regularization by implementing and comparing L2 regularization (weight decay) and L1 regularization (manual term). Compare the results of the two approaches.

3. Optimizer Comparison

Train the Logistic Regression model using SGD and Adam. Compare their convergence and accuracy.

4. Visualization of Weights

Plot the learned weight vectors as 28×28 images. Interpret the resulting visualizations.

5. Confusion Matrix

Compute and plot a confusion matrix using the test predictions. Discuss the common confusions between classes.

6. Noisy Labels

Investigate the effect of noisy training labels by flipping 10% of the labels. Report the resulting change in accuracy.

Part 2 (5%) — CIFAR-10 Classification with a Fully Connected Neural Network

Train a simple feedforward neural network on CIFAR-10 and analyze training dynamics.

1. Data Pipeline

Load CIFAR-10, normalize the data, and create a training set, a validation set, and a test set.

2. Fully Connected Neural Network

Implement a fully connected neural network (FFNN) for CIFAR-10 classification. **Do not use a CNN.** Train the network for a few epochs.

3. Training Dynamics

Plot training loss vs. validation loss, and training accuracy vs. validation accuracy. Interpret the resulting curves.

4. Confusion Matrix

Compute and plot a confusion matrix on the validation set. Discuss the common confusion trends.

5. Misclassification Inspection

Display a grid of at least 16 misclassified validation images, showing the predicted and true labels for each image.

6. Model Modification

Make one deliberate change to the model or training setup — for example hidden size, activation, learning rate, or number of epochs. Re-train the model and update the validation metrics and plots. State what you expect to happen before making the change, and report what happened after the experiment.

Report

Write a 1–2 page PDF report summarizing your findings from both parts of the assignment:

- Findings from the regularization experiments.
- Comparison of SGD and Adam.
- Findings from the noisy-label experiments.
- Training dynamics of the fully connected neural network.
- Confusion trends.
- Effect of your model modification.
- AI-assistance disclosure, if applicable.

Performance Expectations

The following performance targets are used for full marks on the training components. These targets are reachable with the notebook's default FAST_MODE settings.

Model	Ideal validation accuracy	Notes
Logistic regression (MNIST)	$\geq 88\%$	MNIST is close to linearly separable in pixel space, so a linear model does well. The optimizer changes how fast you get there, not where you end up.
Fully connected NN (CIFAR-10)	$\geq 45\%$	One hidden layer of 512 units, ~15 epochs, is enough. Random guessing is 10%.

Keep all other hyperparameters fixed when comparing models or training settings. The learning rate may differ between SGD and Adam; state the values used.

Runtime (FAST_MODE)

The notebook defines a FAST_MODE flag that trains on a random subset of each training set so the assignment runs quickly on Colab. It is **enabled by default**. Keep the same setting for every experiment you compare, and state in your report which setting you used. **We would recommend initially running with fast mode on and then running the notebook with fast mode off and report those numbers.**

Rubric

Each rubric criterion is graded independently. Full marks are awarded when **all** requirements within a criterion are satisfied; otherwise, the criterion receives zero. There are no partial marks within an individual criterion.

Part 1 — MNIST (4%)

Component	Weight	Criteria
Baseline Logistic Regression	0.5%	Logistic Regression is correctly implemented and trained using SGD and Cross-Entropy loss.
Regularization	0.5%	Both L1 and L2 regularization are implemented and compared.
Optimizer Comparison	0.5%	Both SGD and Adam are tested, with convergence and accuracy compared.
Visualization of Weights	0.5%	Weight vectors are plotted as 28×28 images and interpreted.
Confusion Matrix	0.5%	A test-set confusion matrix is correctly computed and common confusions are discussed.
Noisy Labels	0.5%	10% label noise is implemented and the resulting accuracy change is reported.
Part 1 Analysis	1.0%	Findings from the MNIST experiments are summarized and interpreted.

Part 1 Total = 4%

Part 2 — CIFAR-10 (5%)

Component	Weight	Criteria
Data Pipeline	0.25%	CIFAR-10 is loaded, normalized, and split into training, validation, and test sets.
Fully Connected Neural Network	0.5%	A fully connected neural network is correctly implemented and trained without using a CNN.
Training Curves	0.25%	Training and validation loss and accuracy are plotted and interpreted.
Confusion Matrix	0.25%	A validation-set confusion matrix is correctly computed and common confusion trends are discussed.
Misclassification Grid	0.25%	At least 16 misclassified validation images are shown with predicted and true labels.
Model Modification	1.0%	One deliberate change is made; the expected outcome is stated before the experiment and the actual outcome is reported afterwards.
Modified Evaluation	0.5%	Validation metrics and plots are updated for the modified configuration.
Part 2 Analysis	1.0%	Training dynamics, confusion trends, and the effects of the model modifications are analyzed.
Report	1.0%	A 1–2 page report summarizes the required findings from both parts.

Part 2 Total = 5% · Assignment Total = 9%